UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Sc. Engineering Part III Even Semester Exam -2024

CSE 3252: Computer Networks Lab.

For Affiliated Colleges

Roll No: **Time: 03 Hours**

[Solve the marked two]

1. a) Building Ethernet cable (straight) and testing.
   b) Find the IP Address of your PC.
   c) Set the IP address of PC1: 192.168.0.1 and PC2: 192.168.0.2 – ping from PC1 to PC2.
   d) Share a folder in PC1 and access it from PC2.

2. Write a client-server program to implement the following protocol:

PC1 (Client)	PC2 (Server)
Hi	Hello
Date	Server Date
Time	Server Time
IP	Server IP

   - You can use the following method to get the local machine's IP address.
     java.net.InetAddress.getLocalHost().getHostAddress()

3. **Packet Tracer Problem Statement: Inter-VLAN Routing Using Router-on-a-Stick**

   Objective: Design and simulate a small network in Cisco Packet Tracer where two manageable switches and a single router are used to implement inter-VLAN routing.

   Network Setup:

   You are provided with the following devices:
   - **2 Manageable Switches** (Switch1 and Switch2)
   - **1 Router** (Router0)
   - **10 PCs** connected to each switch (total 20 PCs)

   VLAN Configuration:
   - On both **Switch1** and **Switch2**:
     - **Ports Fa0/1 to Fa0/5** should be assigned to **VLAN 101**
     - **Ports Fa0/6 to Fa0/10** should be assigned to **VLAN 102**

   Requirements:
   a. **Configure VLANs** on both switches.
   b. Assign **PCs in ports 1–5** to VLAN 101 and **ports 6–10** to VLAN 102.
   c. Configure **trunk links** between:
      - Between Switch1 and Switch2 (if needed for full connectivity)
   d. Assign proper IP addresses to all PCs based on their VLAN.
   e. **Verify inter-VLAN communication**, i.e., a PC in VLAN 101 should be able to ping a PC in VLAN 102 across switches.

CamScanner